

Chapter 4 – Installing Kali Linux

Learning Objectives

After completing this chapter, you will be able to:

- Boot the Kali Linux Installer.
 - Configure language, region, and keyboard settings.
 - Configure networking and hostname.
 - Create a user account.
 - Partition the virtual hard disk.
 - Install the base operating system.
 - Install the GRUB bootloader.
 - Successfully boot into Kali Linux for the first time.
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4.1 Starting the Virtual Machine

Open **Oracle VirtualBox**.

Select the **Kali Linux** virtual machine.

Click **Start**.

VirtualBox will boot from the Kali Linux ISO that was attached in Chapter 3.

4.2 Kali Boot Menu

After the virtual machine starts, the Kali Linux boot menu appears.

You will see several options.

For beginners, select:

Graphical Install



The graphical installer provides a user-friendly interface.

4.3 Select Language

The installer will ask you to choose your preferred language.

Recommended:

English

Selecting a language determines:

- Installation language
- System language
- Error messages
- Default locale



4.4 Select Location

Choose your country or region.

Example:

Pakistan

This setting configures:

- Time zone
- Regional settings
- Package mirrors

4.5 Configure Keyboard Layout

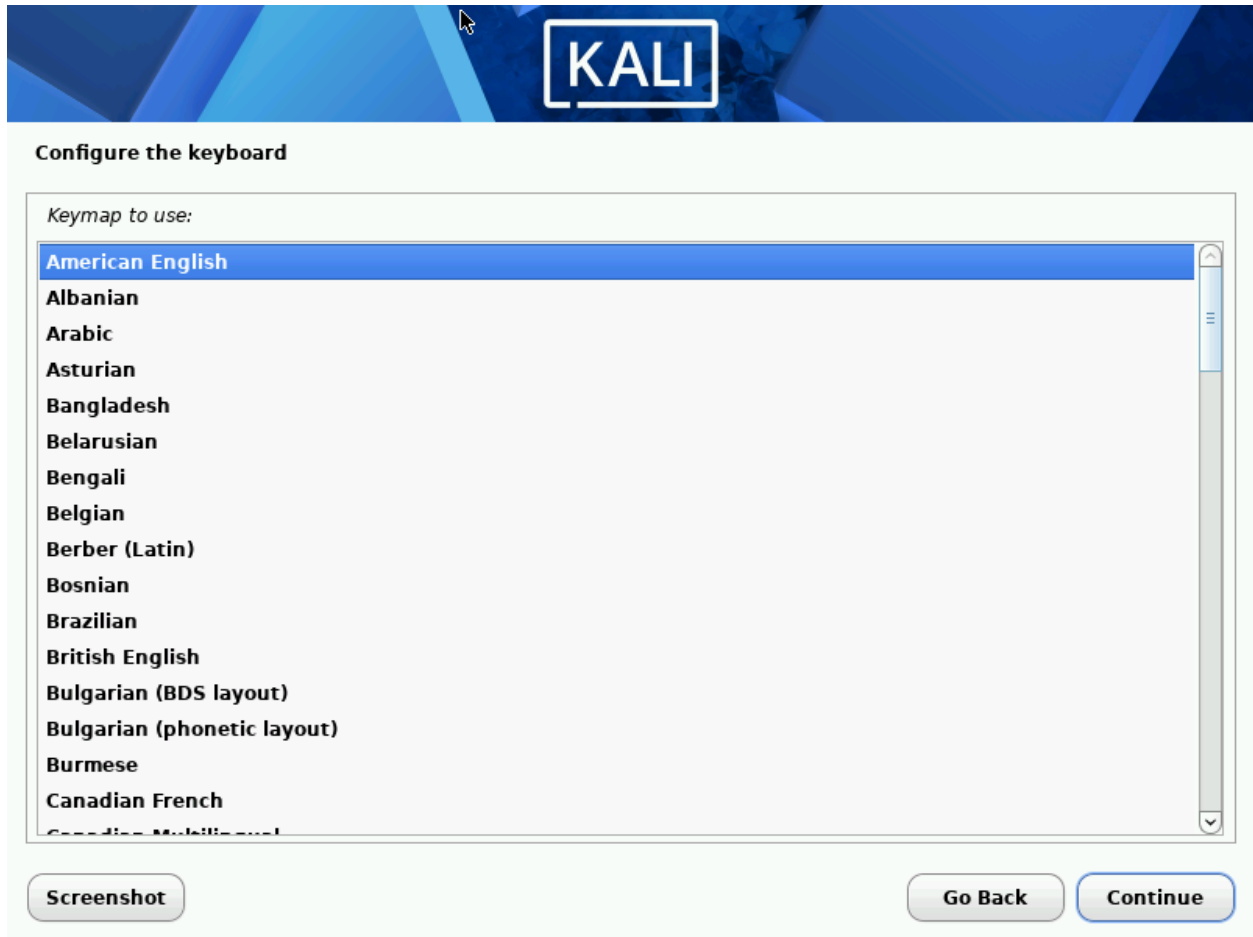
Choose the keyboard layout that matches your physical keyboard.

Most users should select:

American English

or

English (US)



4.6 Detect Hardware

The installer now automatically detects hardware including:

- CPU
- RAM
- Storage

- Network Adapter
- Virtual Hardware

No user interaction is required during this step.

4.7 Configure the Hostname

Every Linux computer has a hostname.

The hostname uniquely identifies the system on a network.

Example:

kali

or

kali-lab



Configure the network

Please enter the hostname for this system.

The hostname is a label that identifies your system. It may be either a single word giving it a unique name within the local network, or a fully qualified domain name (FQDN) which combines the hostname and domain name in one string, separated by periods.

If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make up the hostname yourself.

Hostname:

[Screenshot](#) [Go Back](#) [Continue](#)

Why is a Hostname Important?

Imagine you have multiple systems in your lab:

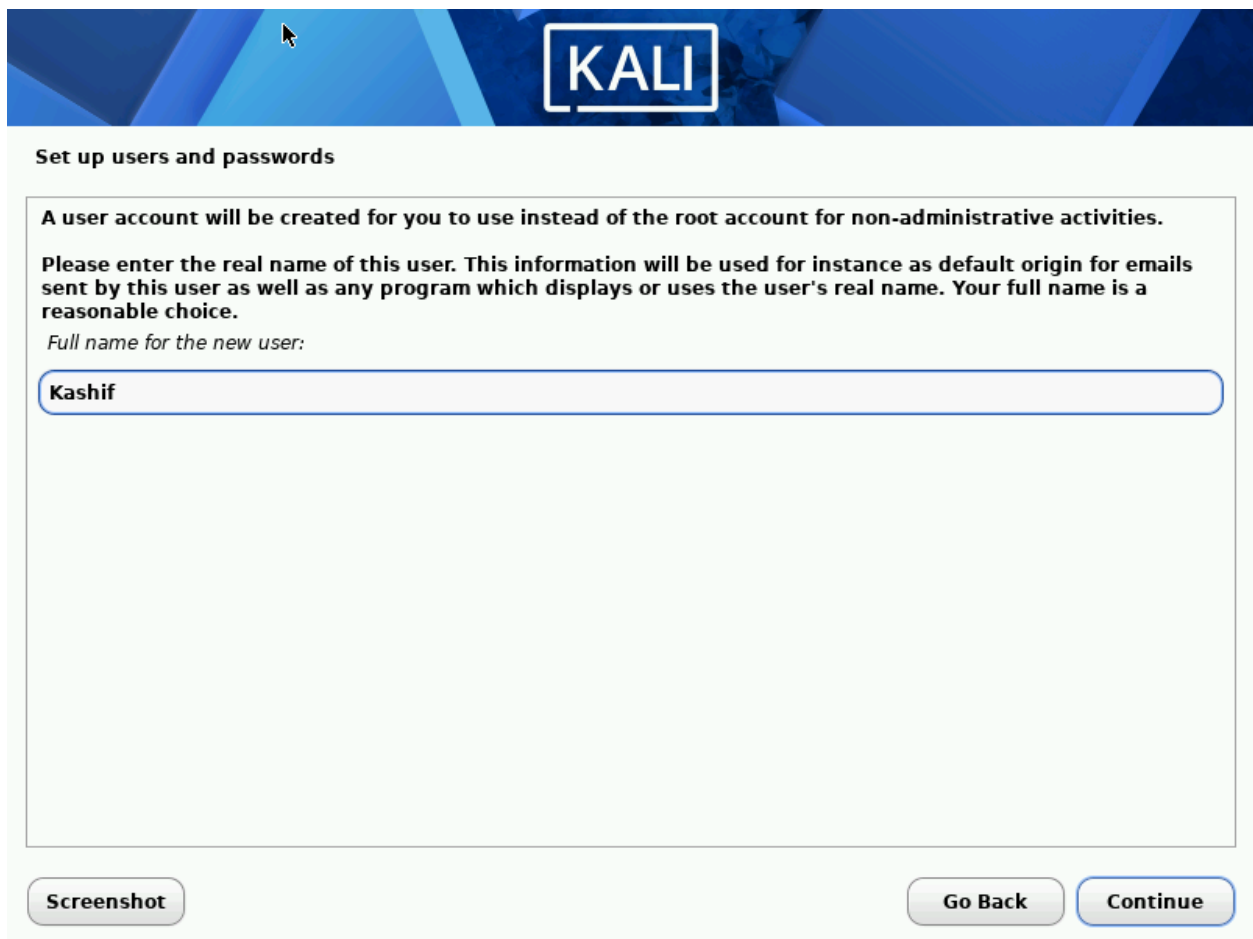
Windows-PC

Ubuntu-Server

Kali

Wazuh

Each hostname helps you identify the correct machine.

A screenshot of the Kali Linux installation process. At the top is a blue banner with the 'KALI' logo. Below it, the text 'Set up users and passwords' is displayed. A message states: 'A user account will be created for you to use instead of the root account for non-administrative activities. Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice. Full name for the new user:'. Below this is a text input field containing the name 'Kashif'. At the bottom of the window are three buttons: 'Screenshot', 'Go Back', and 'Continue'.

4.8 Configure Domain Name (Optional)

For home labs, a domain name is not required.

Leave this field blank.

Click **Continue**.

If you later build an Active Directory environment, this setting can be configured separately.

4.9 Create a User Account

The installer asks you to create your primary user.

Example:

Setting	Example
---------	---------

Full Name	Kashif
-----------	--------

Username	kashif
----------	--------

This account will be used to log in after installation.

4.10 Create a Strong Password

Choose a strong password.

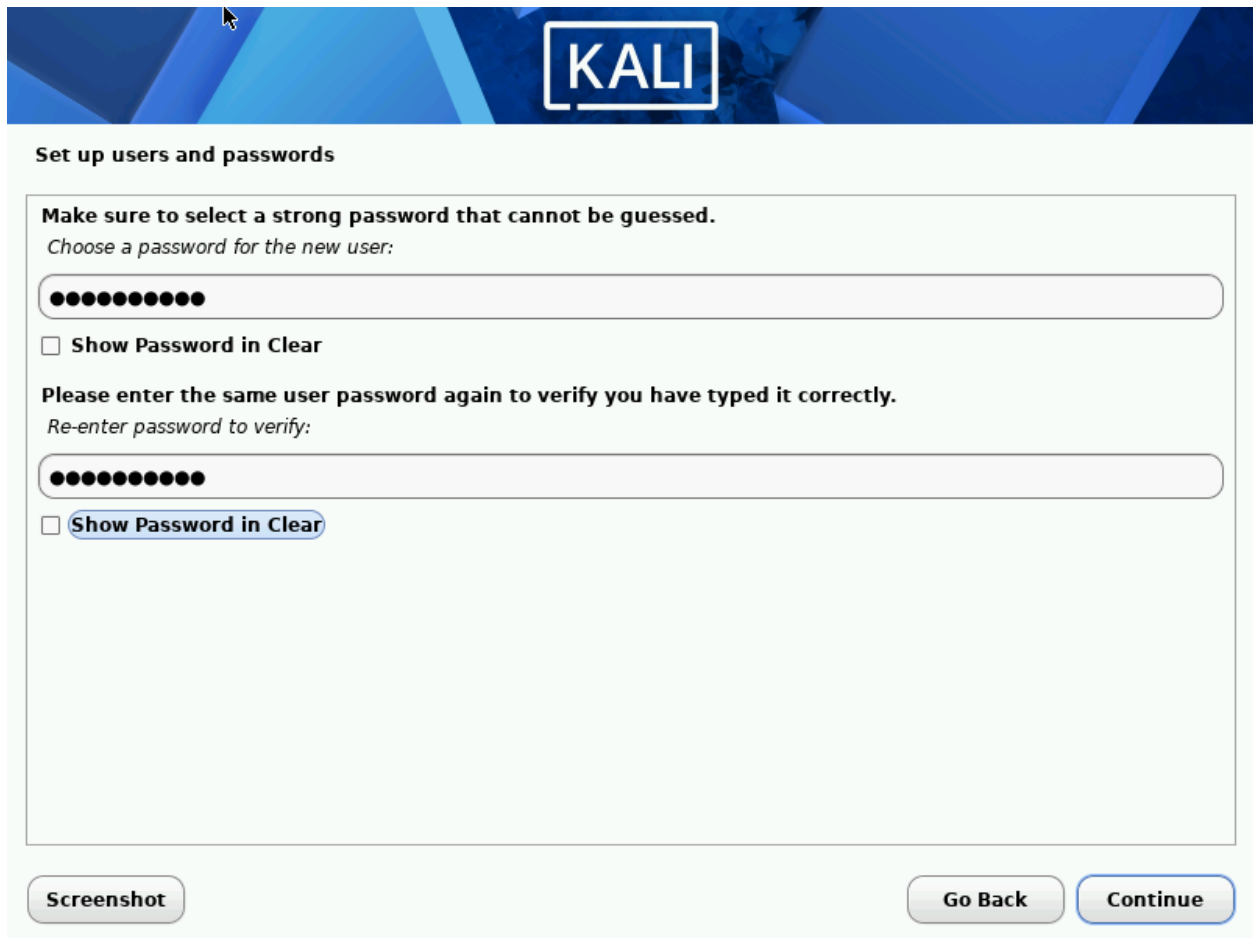
Example characteristics:

- Minimum 12 characters
- Uppercase letters
- Lowercase letters
- Numbers
- Special characters

Example:

P@ssw0rd123!

⚠ Warning: Never use simple passwords such as 123456, password, or your name.



The image shows a screenshot of the Kali Linux installer's 'Set up users and passwords' screen. At the top, there is a blue header with the 'KALI' logo. Below the header, the title 'Set up users and passwords' is displayed. The main content area is a light green box containing instructions: 'Make sure to select a strong password that cannot be guessed.' and 'Choose a password for the new user:'. Below this is a password input field with ten black dots. A checkbox labeled 'Show Password in Clear' is present. The next instruction is 'Please enter the same user password again to verify you have typed it correctly.' followed by 'Re-enter password to verify:'. Below this is another password input field with ten black dots. A second checkbox labeled 'Show Password in Clear' is also present. At the bottom of the green box, there are three buttons: 'Screenshot', 'Go Back', and 'Continue'. The 'Continue' button is highlighted with a blue border.

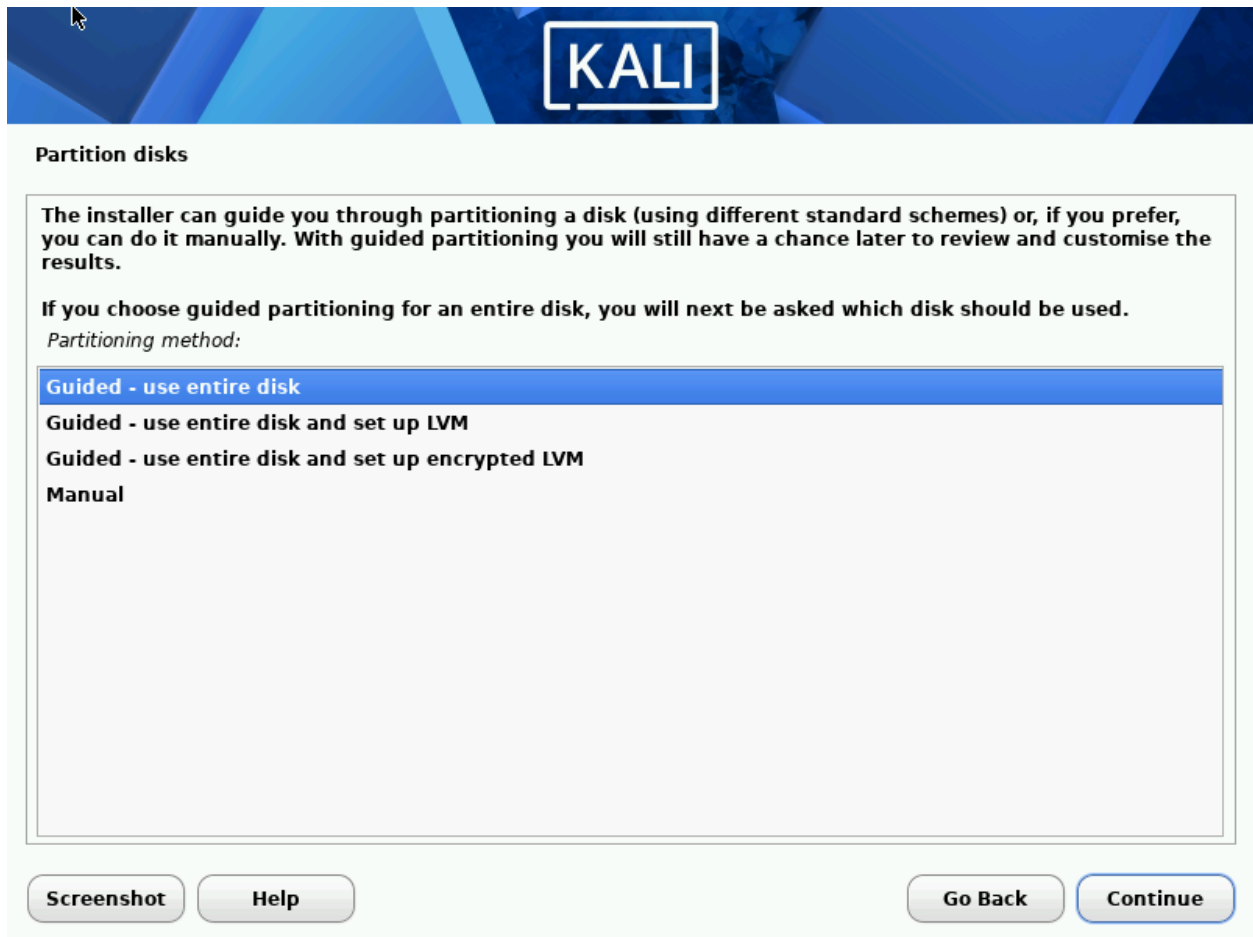
4.11 Partition the Disk

The installer will ask how you want to partition the virtual hard disk.

Since this guide uses a dedicated virtual machine, choose:

Guided – Use Entire Disk

This is the easiest and safest option for beginners.



4.12 Select the Virtual Disk

If only one virtual hard disk exists, it will be displayed automatically.

Example:

80 GB Virtual Disk

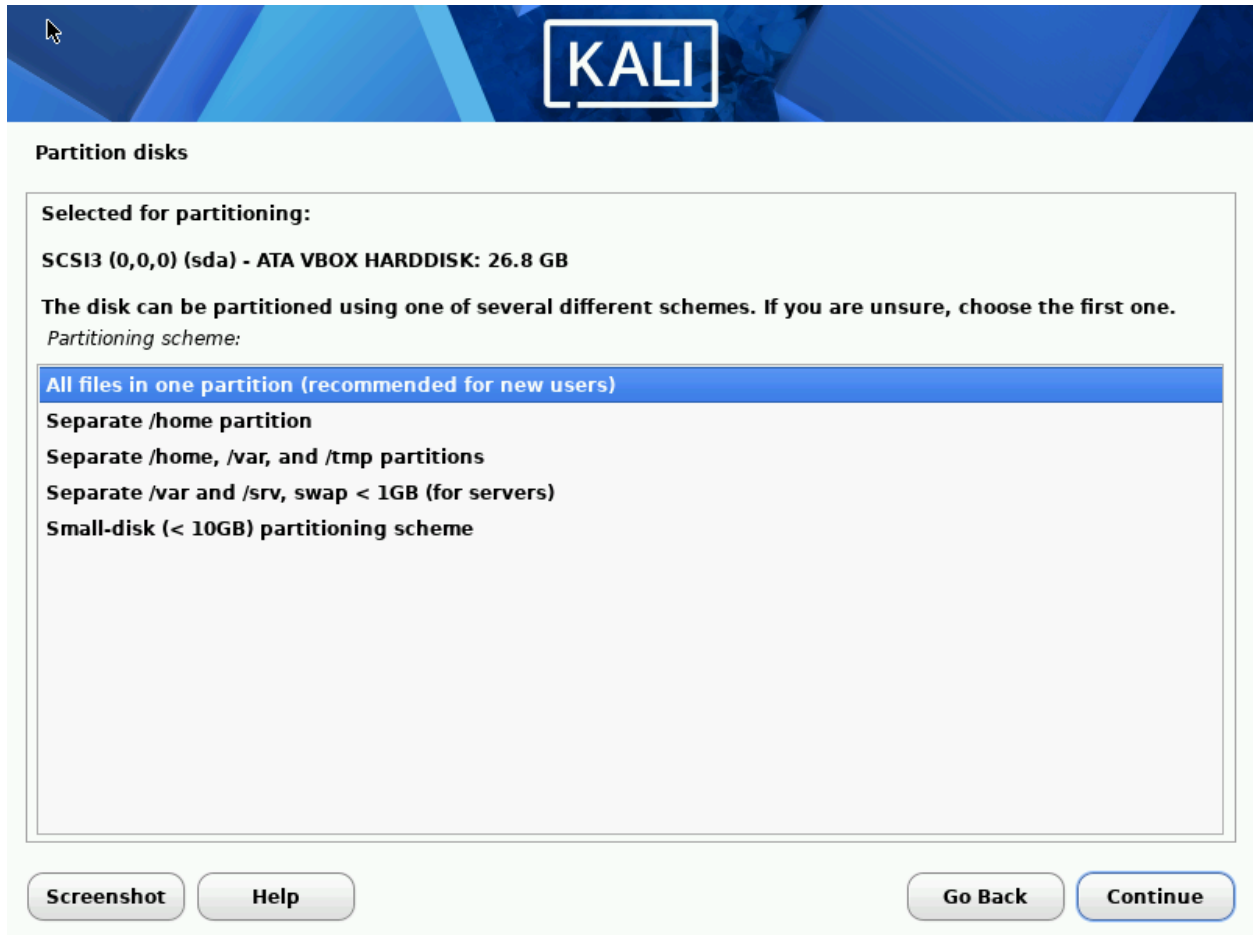
Select it and continue.

4.13 Partitioning Scheme

Choose:

All files in one partition

This layout is suitable for learning environments and home labs.



4.14 Review Partition Changes

The installer displays a summary of the changes.

Example:

Create Root Partition

Format ext4

Write Changes to Disk

Review the information carefully.

If everything looks correct, select:

Finish Partitioning and Write Changes to Disk

4.15 Confirm Disk Changes

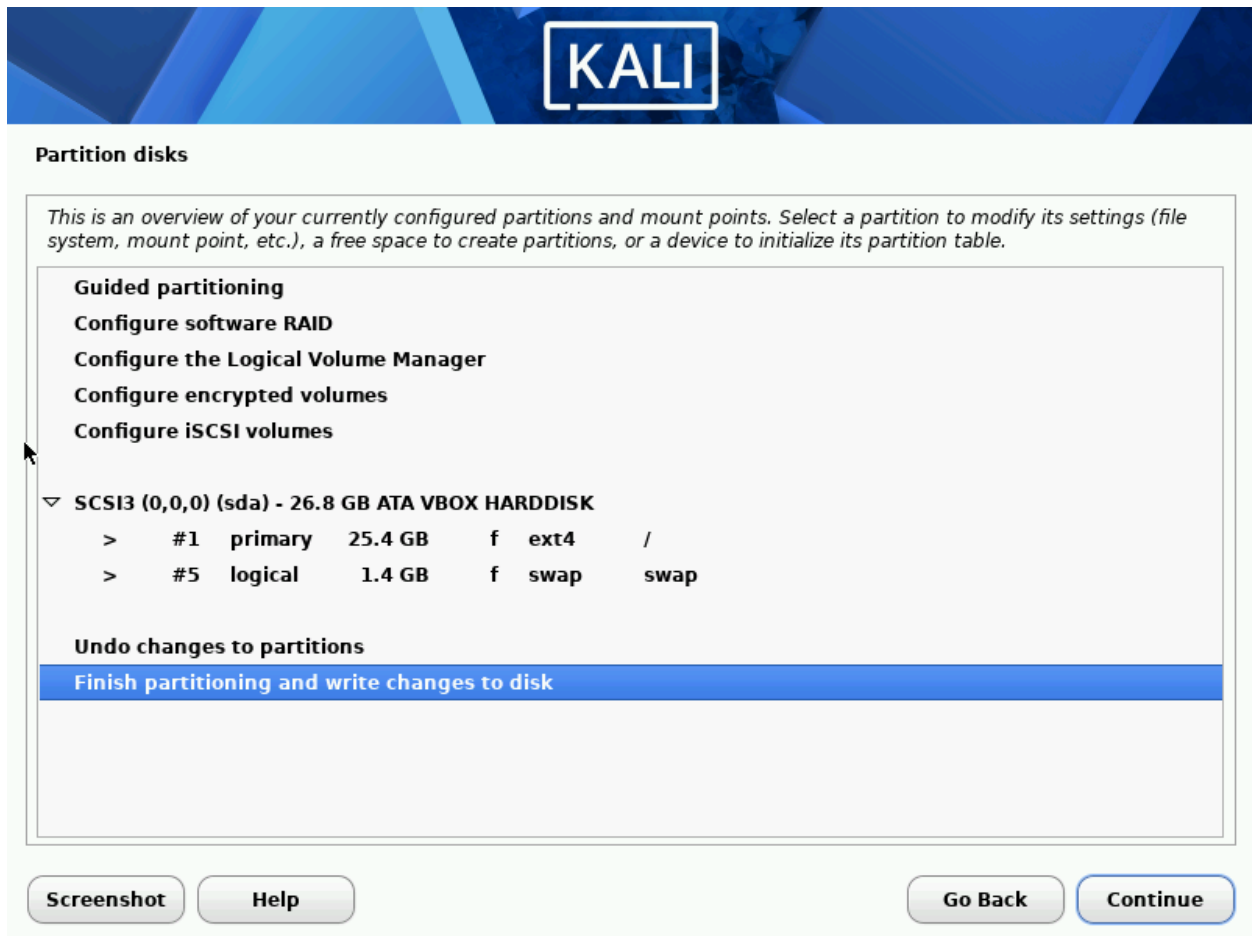
The installer asks for confirmation before formatting the virtual disk.

Select:

Yes

The installation process will begin.

⚠ Warning: This step permanently erases the selected virtual disk. In this guide, only the virtual disk created in Chapter 3 is affected.



4.16 Installing the Base System

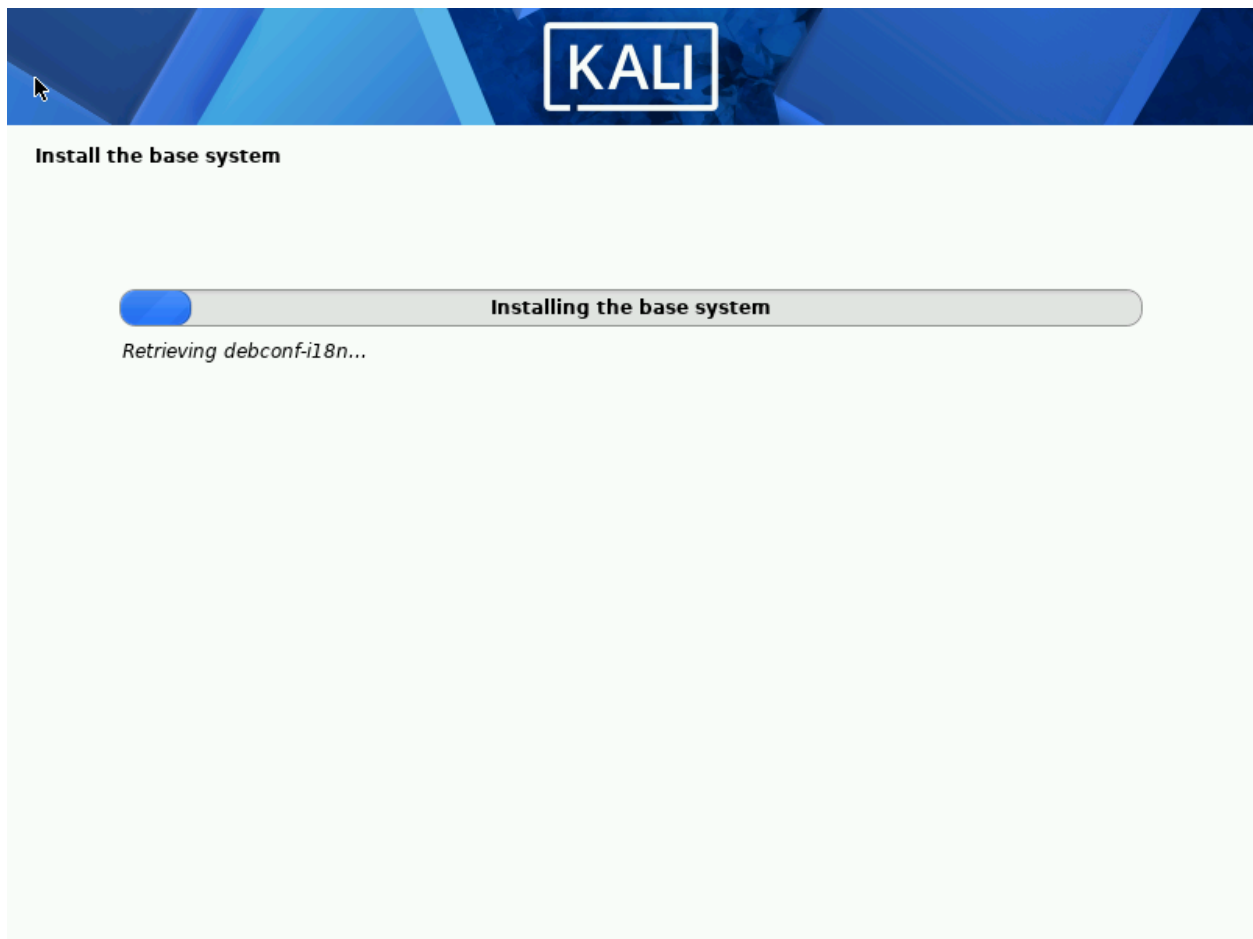
The installer now copies files and installs the Kali Linux operating system.

This process may take several minutes depending on your computer's performance.

During this stage, the installer:

- Copies system files
- Configures packages
- Detects hardware
- Prepares the operating system

No user interaction is required.



4.17 Configure Package Manager

The installer may ask whether to use a network mirror.

Choose:

Yes

Using the official package mirror allows you to download software updates and additional tools after installation.



Software selection

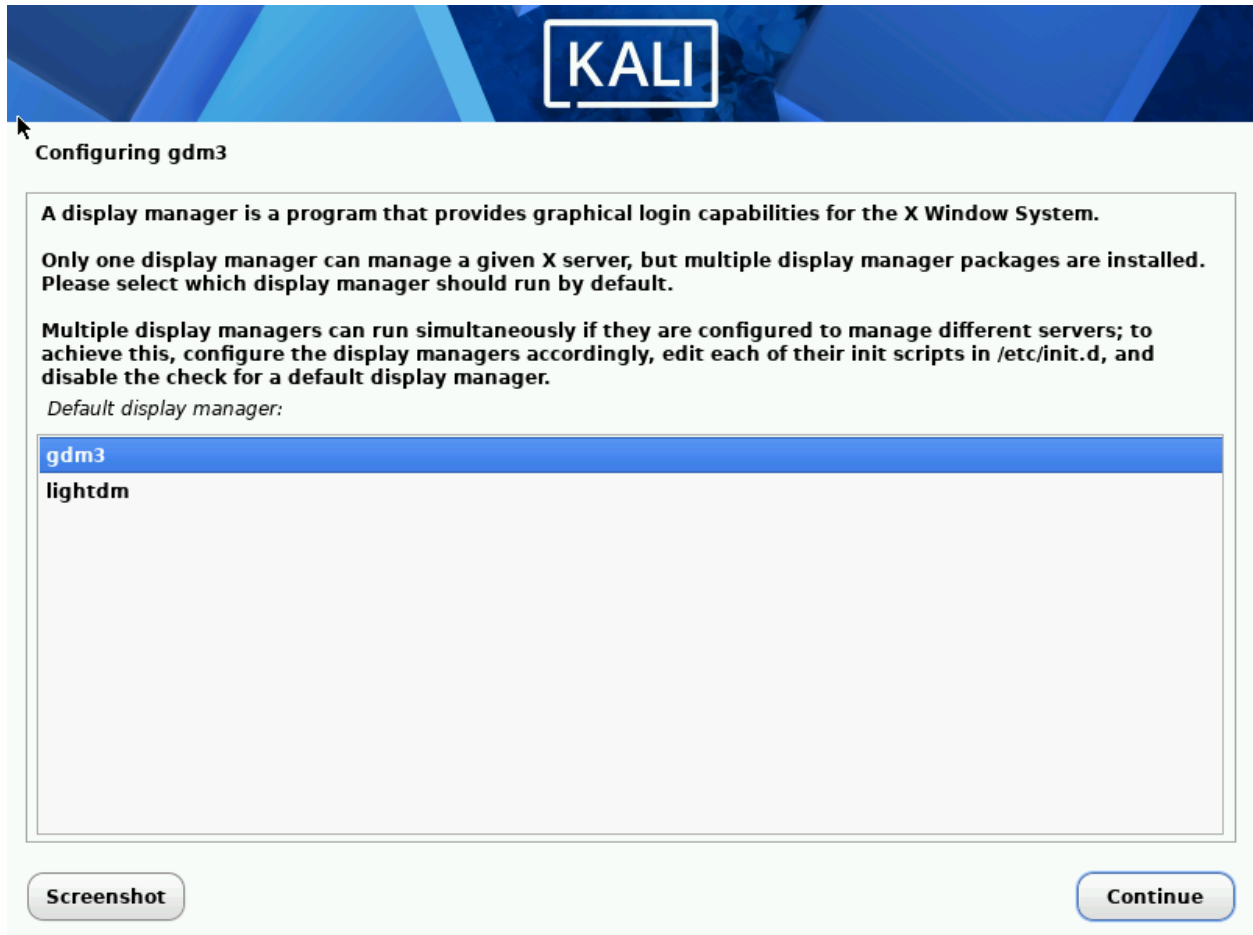
At the moment, only the core of the system is installed. The default selections below will install Kali Linux with its standard desktop environment and the default tools.

You can customize it by choosing a different desktop environment or a different collection of tools.

Choose software to install:

- ☒ Desktop environment [selecting this item has no effect]
- ☒ ... Xfce (Kali's default desktop environment)
- ☒ ... GNOME
- ☐ ... KDE Plasma
- ☒ Collection of tools [selecting this item has no effect]
- ☒ ... top10 -- the 10 most popular tools
- ☒ ... default -- recommended tools (available in the live system)

[Screenshot](#) [Continue](#)



4.18 Install the GRUB Bootloader

GRUB (Grand Unified Bootloader) is responsible for loading the operating system when the virtual machine starts.

Select:

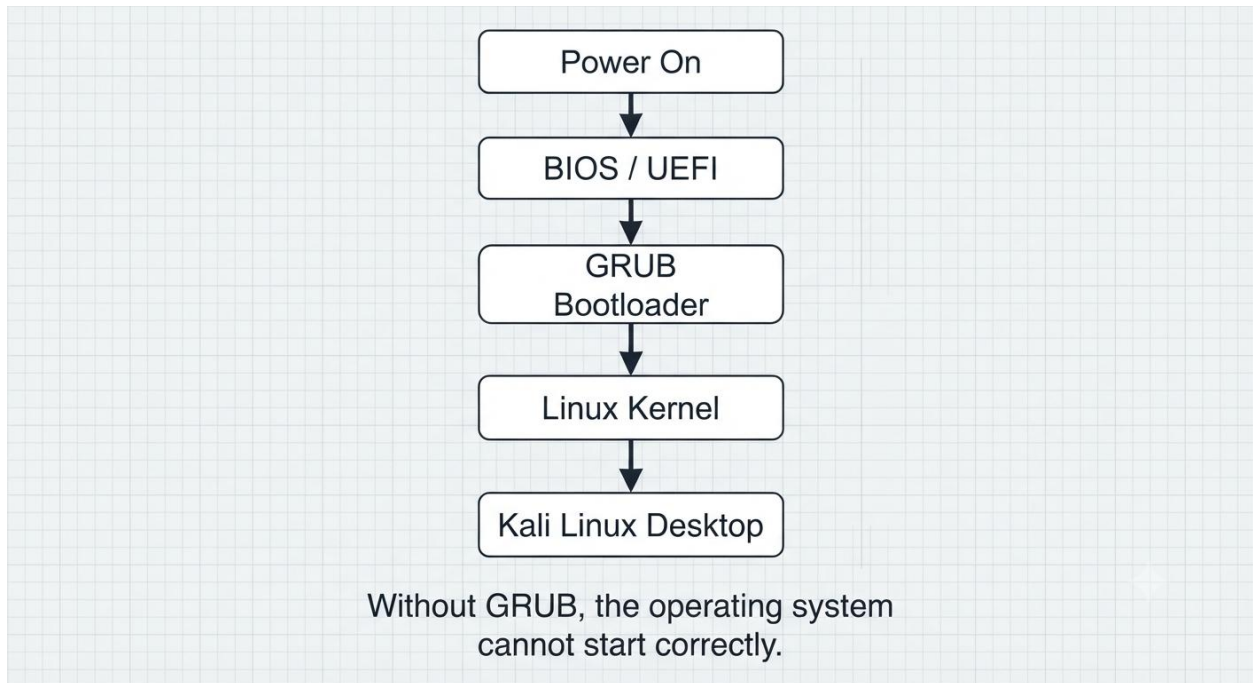
Yes

Install GRUB to the primary virtual hard disk.

What is a Bootloader?

A bootloader starts the operating system.

The boot process is:



Install the GRUB boot loader



Installing GRUB boot loader

Running "grub-install /dev/sda"...

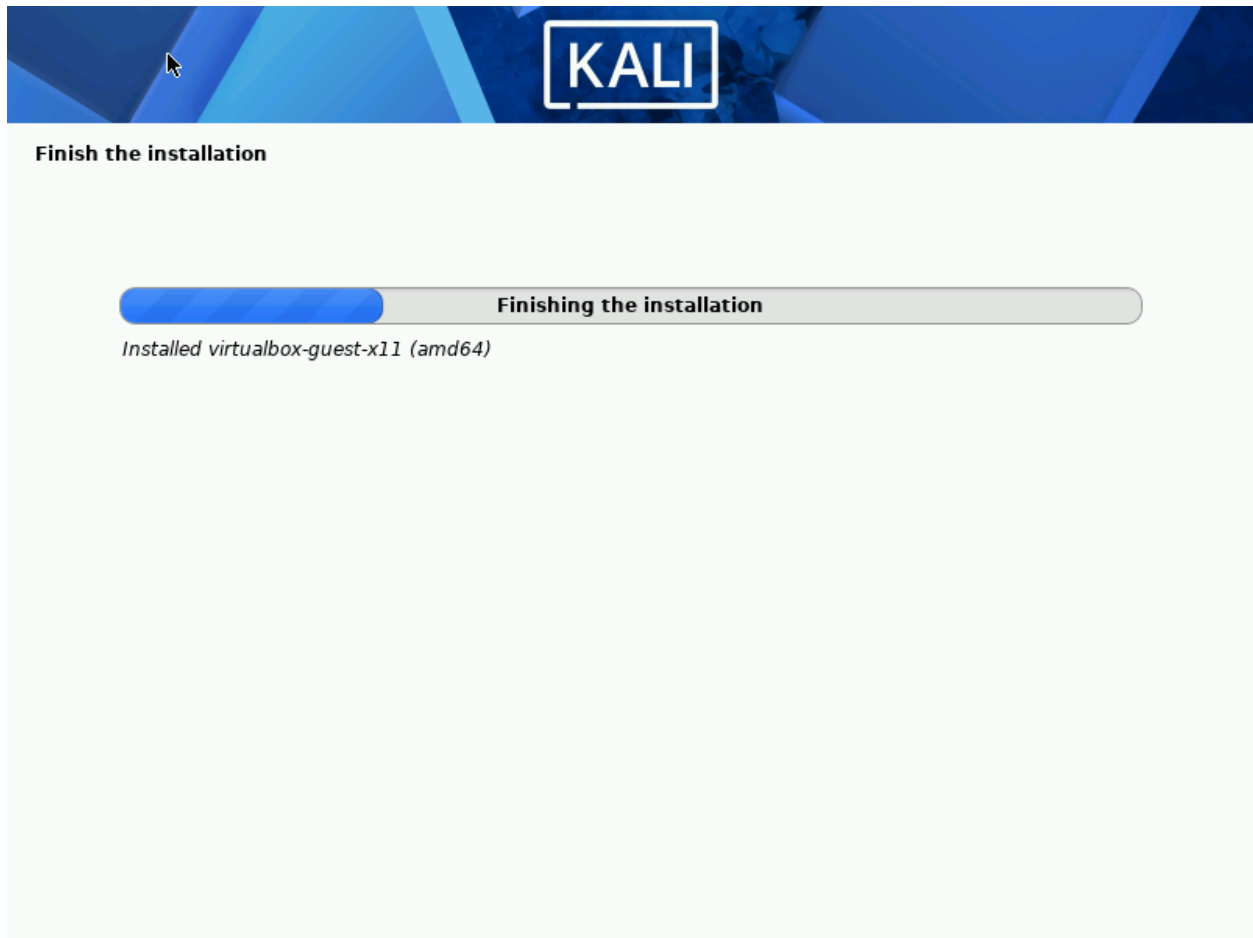
4.19 Finish Installation

Once installation is complete, click:

Continue

The virtual machine will restart automatically.

If prompted, remove the ISO from the virtual optical drive or allow VirtualBox to boot from the installed virtual disk.



4.20 First Boot into Kali Linux

After rebooting, Kali Linux starts from the virtual hard disk.

You will see the login screen.

Enter:

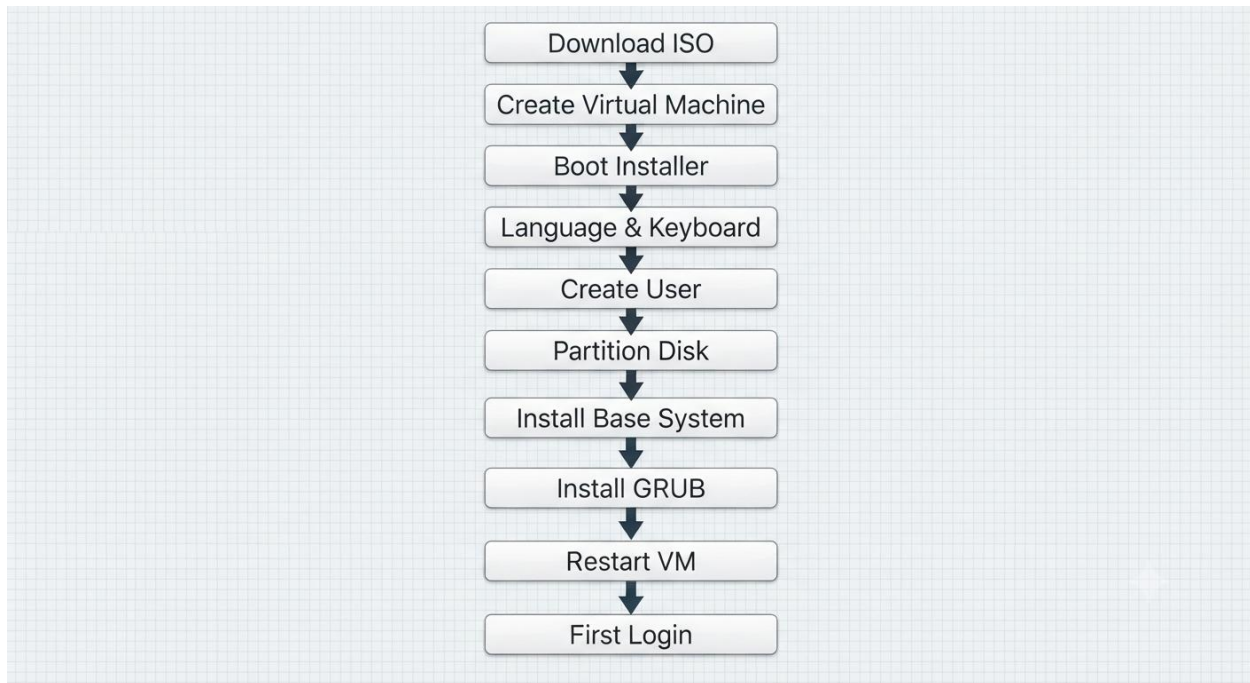
- Username
- Password

created earlier.

If authentication is successful, the Kali Linux desktop loads.



Diagram – Installation Workflow



Best Practices

- Use the **Graphical Install** option unless you have advanced requirements.
- Assign a meaningful hostname for easy identification in your lab.
- Use a strong password for your user account.
- Use **Guided – Use Entire Disk** for dedicated virtual machines.
- Enable the official package mirror to receive updates.
- Verify that GRUB is installed successfully before rebooting.
- Log in and confirm the desktop loads without errors before proceeding.

Chapter Summary

In this chapter, you learned how to:

- Start the Kali Linux installer.
- Configure language, location, and keyboard settings.
- Set a hostname and create a user account.

- Partition the virtual disk using the guided method.
- Install the Kali Linux base system.
- Configure the package manager.
- Install the GRUB bootloader.
- Complete the installation and log in to Kali Linux for the first time.

One-Line Summary

Installing Kali Linux involves configuring system settings, creating a user account, partitioning the virtual disk, installing the operating system, and completing the first successful boot into the desktop environment.
